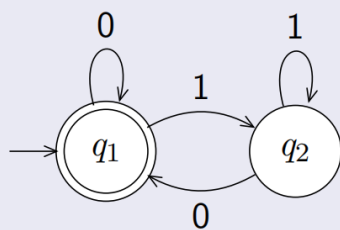


1.给DFA的转移图，写出五元组定义，以及对应的语言。

(出自ppt第三章内容)

Example: DFA M_3

DFA M_3



$$M_3 = (\{q_1, q_2\}, \{0, 1\}, \delta, q_1, \{q_1\})$$

		0	1
δ :	q_1	q_1	q_2
	q_2	q_1	q_2

$L(M_3) = ?$

$$L(M_3) = \{w \mid w \text{ is the empty string } \varepsilon \text{ or ends in a } 0\}$$

What is the relationship between $L(M_2)$ and $L(M_3)$?

2.给出语言画出DFA的状态转移图

(出自作业2)

a. $\{w \mid w \text{ has at least three a's and at least two b's}\}$

3.给出语言画出NFA的状态转移图

(这道题出自教材《计算理论导引第三版》的练习1.7，中文版英文版都是)

1.7 Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts, the alphabet is $\{0,1\}$.

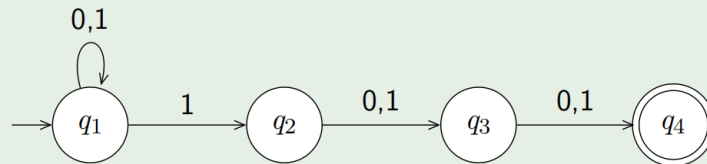
a. The language $\{w \mid w \text{ ends with } 00\}$ with three states

4.NFA转成DFA

(出自ppt，最好使用后面介绍的子集方法来写)

- The language A :
 - {the language consisting of all strings over $\{0, 1\}$ containing a 1 in the third position from the end}
 - e.g., $000100 \in A$, $0011 \notin A$

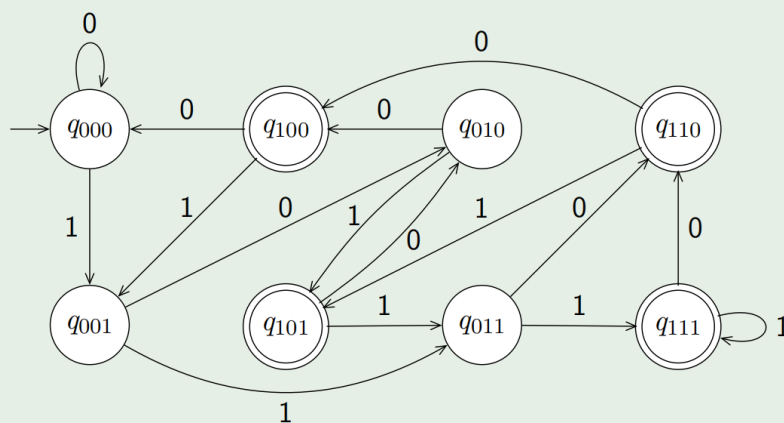
例 (NFA N_2)



$$L(N_2) = A$$

Every NFA can be converted into an equivalent DFA.

例 (The equivalent DFA of NFA N_2)



5. RE转成NFA

(出自作业2)

9. Use the procedure described in Lemma 1.55 to convert the following regular expressions to non-deterministic finite automata.

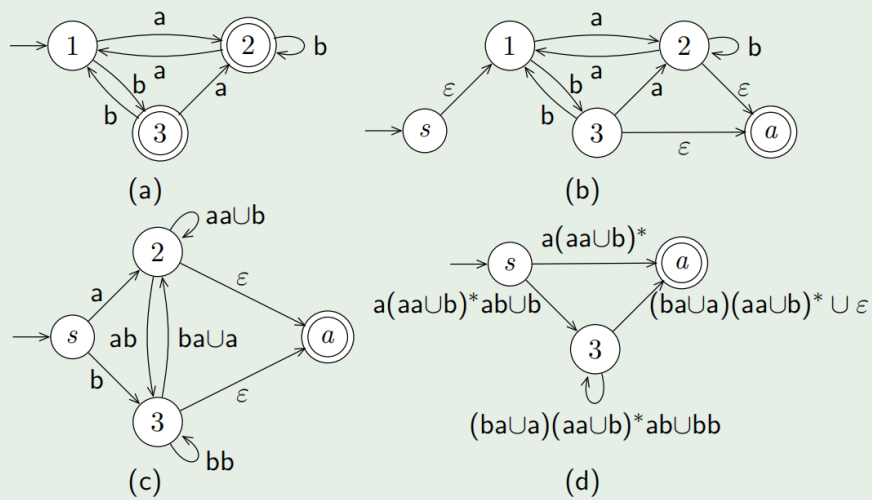
a. $(0 \cup 1)^* 000 (0 \cup 1)^*$

6. DFA转成RE

(出自ppt)

Converting a DFA into a regular expression

例 (Converting a 3-state DFA to an equivalent regular expression)



Converting a DFA into a regular expression

例

$(a(aaUb)^*abUb)((baUa)(aaUb)^*abUbb)^*((baUa)(aaUb)^* \cup \epsilon)Ua(aaUb)^*$



(d)

7.使用泵引理证明不是正则语言

(出自ppt)

例

Let $D = \{1^{n^2} \mid n \geq 0\}$. Use the pumping lemma to prove that D is not regular.

Proof. (The proof is by contradiction.)

- Assume to the contrary that D is regular.
- Let p be the pumping length given by the pumping lemma.
- Let s to be the string 1^{p^2} .
- Because s is a member of D and s has length at least p , the pumping lemma guarantees that s can be split into three pieces, $s = xyz$, where for any $i \geq 0$ the string xy^iz is in D .

We show that this outcome is impossible.

Let $D = \{1^{n^2} \mid n \geq 0\}$. Use the pumping lemma to prove that D is not regular.

Now consider the two strings xyz and xy^2z .

- By condition 3 of the pumping lemma, $|xy| \leq p$ and thus $|y| \leq p$.
- We have $|xyz| = p^2$ and so $|xy^2z| \leq p^2 + p$.
- But $p^2 + p < p^2 + 2p + 1 = (p + 1)^2$.
- Condition 2 implies that $|y| > 0$ and so $|xy^2z| > p^2$.
- Therefore, $p^2 < |xy^2z| < (p + 1)^2$. Hence this length cannot be a perfect square itself.
- So we arrive at the contradiction $xy^2z \notin D$ and conclude that D is not regular.

8.RG转成FA

(出自ppt)

例 (例)

给出正则文法 G_2 如下:

$$S \rightarrow 0A$$

$$A \rightarrow 1A \mid B$$

$$B \rightarrow 0 \mid \varepsilon$$

根据定理给出的方法, 构造对应的 FA。

板书

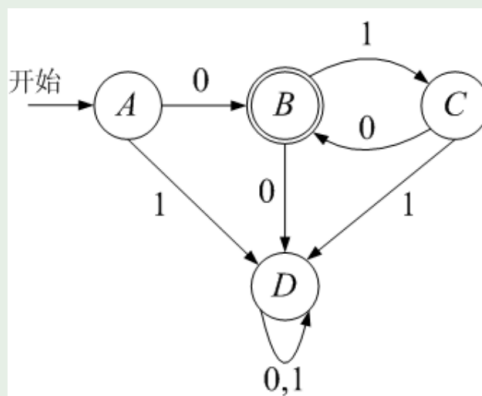
9.DFA转成RG

这个我没有找到原题，但是比ppt上的简单。我记得是把C改成了只进不出的，然后D删除了，A，B不变

RG 和 FA 的关系

例 (例)

给出一个 DFA，如下：



根据定理的构造方法，给出对应的正则文法。**板书**

10.CFG转为乔姆斯基范式

(出自作业3)

4. Convert the following CFG into an equivalent CFG in Chomsky normal form, using the procedure given in Theorem 2.9.

$$\begin{aligned} A &\rightarrow BAB \mid B \mid \varepsilon \\ B &\rightarrow 00 \mid \varepsilon \end{aligned}$$

11.CFG转为PDA

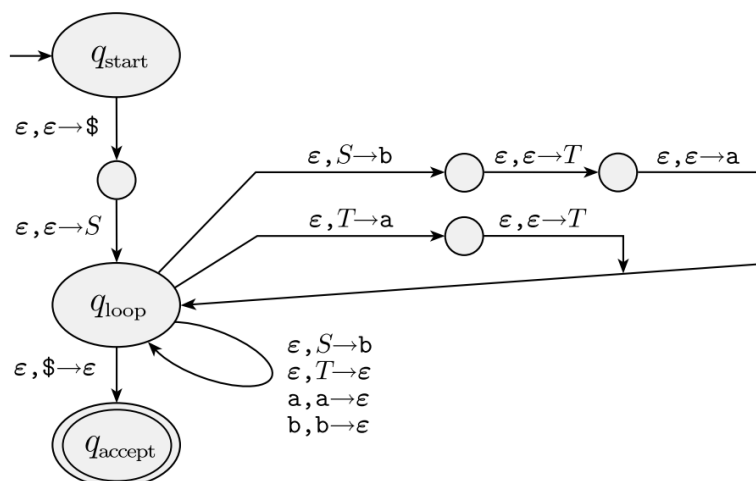
(这个题目也是教材里面的，中文版英文版的序号可能不一样)

EXAMPLE 2.25

We use the procedure developed in Lemma 2.21 to construct a PDA P_1 from the following CFG G .

$$\begin{aligned} S &\rightarrow aTb \mid b \\ T &\rightarrow Ta \mid \varepsilon \end{aligned}$$

The transition function is shown in the following diagram.

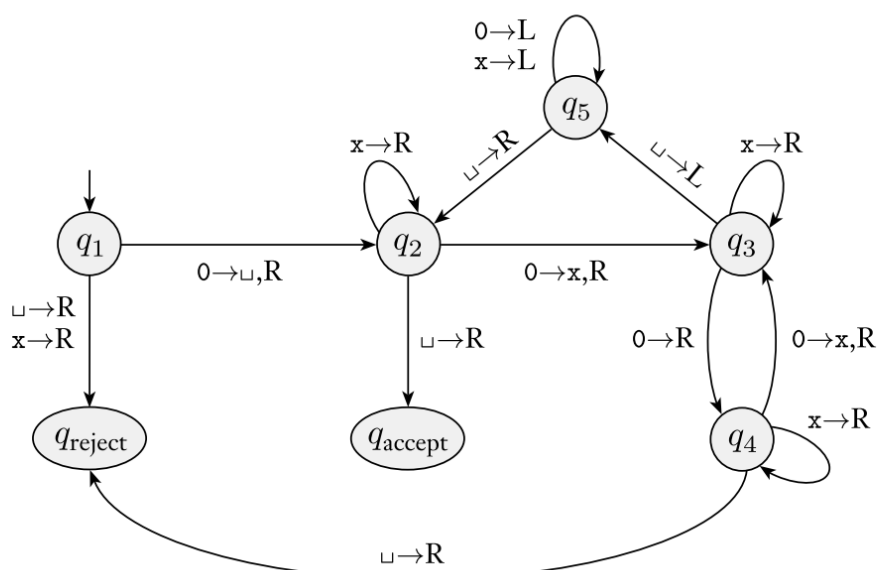


12. 给出图灵机转移图，写出读入某个串的格局的转移

(教材里面的图，第三章，作业里面好像也有)

EXAMPLE 3.7

Here we describe a Turing machine (TM) M_2 that decides $A = \{0^{2^n} \mid n \geq 0\}$, the language consisting of all strings of 0s whose length is a power of 2.



$q_1 0000$	$\sqcup q_5 x 0 x \sqcup$	$\sqcup x q_5 x x \sqcup$
$\sqcup q_2 000$	$q_5 \sqcup x 0 x \sqcup$	$\sqcup q_5 x x x \sqcup$
$\sqcup x q_3 00$	$\sqcup q_2 x 0 x \sqcup$	$q_5 \sqcup x x x \sqcup$
$\sqcup x 0 q_4 0$	$\sqcup x q_2 0 x \sqcup$	$\sqcup q_2 x x x \sqcup$
$\sqcup x 0 x q_3 \sqcup$	$\sqcup x x q_3 x \sqcup$	$\sqcup x q_2 x x \sqcup$
$\sqcup x 0 q_5 x \sqcup$	$\sqcup x x x q_3 \sqcup$	$\sqcup x x q_2 x \sqcup$
$\sqcup x q_5 0 x \sqcup$	$\sqcup x x q_5 x \sqcup$	$\sqcup x x x q_2 \sqcup$
		$\sqcup x x x \sqcup q_{\text{accept}}$

13.若A是正则语言，则A^R也是正则语言

(出自作业2)

7. For any string $w = a_1 a_2 \cdots a_n$, the reverse of w , written $w^{\mathcal{R}}$, is the string w in reverse order, $a_n \cdots a_2 a_1$. For any language A , let $A^{\mathcal{R}} = \{w^{\mathcal{R}} \mid w \in A\}$. Prove that if A is regular, so is $A^{\mathcal{R}}$.